

Ice Core

Season Science Lead • Vestri Dome

10 DAYS • 41 STOPS • GRADE 6 • ICECORE_MS

THE OPENING CARD

Vestri Dome is a drilling camp on the polar plateau: six huts up on legs, a trench with a tower over it, and two and a half kilometres of ice already pulled out of the ground. None of it can be let above minus twenty. You are the season science lead, which means the climate record this station sends home is the one you signed. A camp four hundred kilometres away has published a record that disagrees with yours. The aircraft that takes everybody off the ice arrives in fifteen days. Whatever has not been measured by then is not measured until the next crew flies in, two years from now.

What the plateau has been writing down

BEFORE THE DAY — GREET

Nine people, one season, and the plane leaves in March

Everybody on this plateau is here for the same short season, and the drill runs on all of them agreeing. Put a name to most of them before the first core comes up. It is a long winter to spend not knowing who is who.

PLAN CARD 76W

First morning of the season, and the drill has not turned yet. Nadia Brandt has a fortnight of stake readings and a fresh pit wall. Marit Halvorsen wants the season's snowfall figure before the plan is written. Today you settle how much snow a year adds, and why the hole was put here rather than forty kilometres downhill. Get the snowfall wrong and every age below it is wrong the same way, quietly, for three kilometres.

BRIEFING 13W

A new season opens on a core that is already 2000 metres deep.

WHAT YOU DECIDE 15W

Establish what this site records, how fast it records it, and why anybody drilled here.

1.1 How thick a year is here

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · BALLPARK

SCENE 27W

Brandt has the pit wall photographed. Last summer's surface is marked thirty-two centimetres down. The snow above it weighs about a third of what solid ice would.

GUIDE 65W

Five numbers, and two of them belong elsewhere. How heavy water is, and the days of stake readings. Ask of each whether this sum needs it. Snow and ice are the same stuff at different weights, so a thickness of snow is not a thickness of record. Only ice lets two sites be compared. This one gets about a tenth of what the coast gets.

ASK 9W

How much ice will this year's snow turn into?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

snow turns into ice by being squashed under the snow that falls on top of it · density is how much a given amount of something weighs

VERDICT 74W

Snow and ice are the same stuff at different weights, so a thickness of snow is not a thickness of record. Scale it by how heavy the snow is against solid ice, and you get what it will become once the air is squeezed out. That is the only form in which two places can be compared. Here a year is about twelve centimetres of ice. A site by the coast gets a metre.

TAKEAWAY 15W

A year's snowfall means nothing until it is written as the ice it will become.

1.2 Why the hole is on the dome

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · CHOICE

SCENE 35W

Pär Lindqvist, the drill engineer, has the hole down at two thousand metres. On the wall is a map with three abandoned sites, all of them on slopes. Vestri Dome is the high point here.

GUIDE 55W

Four things a dome might buy, and three of them are about cold, thick or flat. Ask of each whether it is about where the ice came from. An ice sheet spreads out from its high points. On a slope, the ice deep down arrived from somewhere else. Domes pay for that with thin layers.

ASK 14W

What does drilling on the dome buy that a site on a slope cannot?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

ice spreads outwards from the high points of an ice sheet

VERDICT 77W

An ice sheet spreads outwards from its high points, like very slow treacle. On a slope, the ice at a given depth came from somewhere uphill. So a core there mixes one place's weather with a long journey, and nothing afterwards can separate them. Under a dome the ice goes almost straight down, so the column under the drill is very nearly the snow that fell on it. Domes charge for that. High, cold, dry, thin layers.

TAKEAWAY 18W

A core is a column of what fell above it only where the ice has not travelled sideways.

1.3 What makes one year different from the next

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · PROTOCOL

SCENE 33W

A metre of core from sixty metres down is on the light table. Ines Okonkwo, the chronology lead, has four things marked on it and wants each one explained before it is cut.

GUIDE 50W

Four things marked on one metre of core, and four things that made them. Pair them by asking what each is made of. Crystals? Acid? Dust grains? Bubbles? Then ask which season it belongs to. No one marker lasts the whole core, so a year is where several signals agree.

ASK 11W

Match each thing in the ice to what put it there.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

snow that falls in summer sits in different conditions from snow that falls in winter

VERDICT 75W

No single marker survives all the way down a core. So a year is counted where several different signals line up. Coarse crystals grow on a summer surface left out in the open. A spike in electricity is acid, and volcanic acid arrives in one season. Dust blows in from far away and peaks with the winds there. The banding is bubbles changing as each layer was squashed. Three of four agreeing is a year.

TAKEAWAY 15W

A year is counted from several signals that agree, not from one that looks convincing.

END OF THE DAY 14W

A record is only as good as the rate at which it was written.

Three kilometres of one attempt

BEFORE THE DAY — TRIAL

Walk the camp along the flag line

The camp is four modules on legs, joined by a line of flags. Walk it once in daylight. When the wind gets up you cannot see the next module, and the flags are the only thing that says which way is home.

PLAN CARD 72W

Second day, and the drill is going back down after a night of repairs. Lindqvist has three short runs on the sheet from last season's brittle ice. Sigrid Holt has a section that arrived with no depth mark on it. Today you decide how the brittle run is made, and what happens to a section nobody can place. Ice gets one reading. What breaks on the way up cannot be drilled again.

BRIEFING 17W

The brittle zone starts at 540 metres and the run sheet has 3 short runs on it.

WHAT YOU DECIDE 16W

Get the core to the surface and onto the line without changing what it can say.

2.1 The run through the brittle zone

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · SEQUENCE

SCENE 32W

The barrel is at seven hundred and eighty metres, in the depth band where the ice comes up cracked. The winch, the fluid and the storage are all set for ordinary ice.

GUIDE 58W

All five steps will happen, so ask what each one buys the next. Down in that band the bubbles are squeezed harder than the surface can hold. Ask of each step whether it softens the shock, lets the pressure ease, or saves something about to be lost. One of them throws away the depths if it comes first.

ASK 13W

Put the steps in the order that brings a cracking section up whole.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

air trapped deep in the ice is squeezed to the pressure of the ice around it

VERDICT 74W

Deep down, the bubbles are squeezed far harder than the air at the surface. Bring that ice up quickly and it cracks itself apart. Each step here buys the next one. A slow lift is less of a shock. Resting at hole temperature lets the pressure even out. Writing down the breaks fixes which depth each piece came from, while somebody still knows. Cutting first destroys that, and cuts ice that is still moving.

TAKEAWAY 12W

Ice brought up badly is a shorter record, not a cheaper one.

2.2 The metre with no label

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CHOICE

SCENE 32W

A section has come to the line with its depth mark broken. Holt can place it to within about four metres from the run sheet. The isotope team has asked for it.

GUIDE 62W

Four options for a section nobody can place better than four metres. Ask of each what the depth is for. It is what turns a measurement into a statement about a year, and four metres here is about thirty years. Some work never uses depth at all. And the doubt has to stay with the sample, or somebody later will not know.

ASK 7W

What should the section be used for?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a measurement belongs to the depth it came from, and to no other

VERDICT 80W

A depth is what turns a measurement into a statement about a year. Four metres here is thirty-odd years, so an isotope value off this section cannot be put anywhere in the record. The ice is still perfectly good for anything a depth does not enter. How heavy it is. What the crystals look like. How the drill is performing. What matters is that the doubt travels with the sample. The danger is somebody later who never hears about it.

TAKEAWAY 12W

A sample with no depth is a measurement of nothing in particular.

2.3 Clean until it was handled

COLD LABORATORY · A ROOM · CHOICE

SCENE 46W

Elena Cruz has a salt result four times the run before it. The blank run before it was clean, and the blank after it is high. The outer centimetre was trimmed off as it should be, and the same year in the second core is ordinary.

GUIDE 59W

Four options, and the clean readings do the work. The blank before was fine and the blank after is high. The trim was done properly. The same year in the second core is ordinary. Ask of each option how many of those it fits. What names the source is not the loud number but which quiet ones stayed quiet.

ASK 19W

The blank before the sample was clean, and the blank after it is high. Where did the salt join?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a blank goes through the whole process with no sample in it, so it measures the process

VERDICT 74W

The blank before the sample is normal, and the blank after it is high. That puts the salt inside the run rather than inside the ice. The trim was done properly, so the outside of the section is out. The second core holds the same year with no salt in it, so that year's weather is out too. What names the source is not the loud number. It is which quiet ones stayed quiet.

TAKEAWAY 15W

A blank measures the process, so anything it shows was never evidence about the ice.

2.4 The level that holds the hole open

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · HOLD

SCENE 37W

The hole is three kilometres deep and the ice around it is trying to close it. The fluid inside holds it open. The level in the tower has to stay where it belongs while the drill runs.

GUIDE 44W

Hold the level inside the band on the gauge. The band gets narrower as the hole gets deeper. Each run of the winch changes the balance and keeps changing it, so set the pump to match rather than waiting for the gauge to move.

ASK 8W

Hold the fluid level while the drill runs.

BEHIND THE BUTTON 57W

Why the hole needs holding open. Ice this deep is under enough pressure to flow, and it slowly squeezes a hole shut. Fluid pushing back from inside is what stops it.

Why too much is bad too. Push too hard and the wall cracks and swallows fluid. That costs the hole and spoils the record around it.

TAKES AS READ

a column of liquid pushes harder the deeper it is

VERDICT 139W

Ice three kilometres down is under enough pressure to flow, so a hole through it closes on its own. What holds it open is the fluid inside, pushing back. Too little and the walls creep in. Too much and the wall cracks and swallows fluid, which costs the hole and spoils the record near it. Everything the drilling does upsets that balance and keeps upsetting it. Pulling the drill up takes away what it was displacing, for as long as the winch runs. Chips come up with fluid on them. Cold fluid off the reel is heavier than what is in the hole and keeps sinking into it. So the pump is set to match, not tapped. The band gets narrower with depth because the gap between closing the hole and cracking it gets smaller the deeper you go.

TAKEAWAY 15W

A hole is held open by a balance, and somebody has to keep making it.

END OF THE DAY 11W

Everything the core can say is decided before anybody measures anything.

Counting to 2000

BEFORE THE DAY — FOLLOW

Sorensen walks a core from the trench to the freezer

A core is cut, logged, cut again and packed, and every one of those steps happens somewhere different. Sorensen walks one all the way through. Stay with her, and watch where the core is out of the cold.

PLAN CARD 74W

Wednesday, and the counting is holding everything else up. Okonkwo has counted by eye to eleven hundred and eighty metres, and the layers get thinner with every hundred. Halvorsen wants the depth and age scale before the aircraft comes. Today you find where counting stops being possible, and choose which yearly signal to count with. Below where the layers close up, every age comes from a model, and a model is not a count.

BRIEFING 16W

Okonkwo needs the depth at which the layer thickness drops below what the line can resolve.

WHAT YOU DECIDE 14W

Establish how a year is counted, and find the depth where counting stops working.

3.1 One year proves nothing

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · BALLPARK

SCENE 29W

There are thirty years of yearly temperatures from the top of the core. The summary wants one number, and one year can swing more than a degree either way.

GUIDE 53W

Four numbers, and two belong to other questions. How much one year swings, and the years overlapping the mast. Ask which two an average needs. And note what an average does not do. It does not make one year's swing go away. It only says where the middle of thirty of them sits.

ASK 9W

What is the average temperature across the thirty years?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

averaging many years lets the random part of each one cancel out

VERDICT 72W

Every year comes out a little warmer or colder than the run of them, because weather is noisy. Add thirty years and divide by thirty, and those swings cancel out. What is left is much closer to the climate of the period than any one year is. It is also why nobody should read a single warm year as a change. One year here swings more than the difference being argued about.

TAKEAWAY 14W

One year is weather. An average of many years is closer to a climate.

3.2 Which marker survives the depth

COLD LABORATORY · A PERSON · CHOICE

SCENE 26W

Yuki Tanabe has four yearly signals measured on the same stretch of core, at fourteen hundred metres. A year down there is about twenty-six millimetres thick.

GUIDE 53W

Four yearly signals, and a year down there is thinner than a fingernail. Ask of each whether the thing being measured can move through the snow. Water moves through the spaces for decades, and that blurs anything too thin. Salt and dust grains do not move. Two of these fade for other reasons.

ASK 14W

At fourteen hundred metres, which signal can still separate one year from the next?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

water can move slowly through the connected air spaces in packed snow

VERDICT 70W

Water keeps moving through the open spaces in packed snow for decades after it lands. That movement smooths the isotope record, and it smooths away anything thinner than the distance it travels. By fourteen hundred metres a year is far thinner than that. Salt and dust do not move. They are grains and specks sitting where they fell, so their yearly pattern survives as deep as anybody can measure it.

TAKEAWAY 17W

A signal is only useful for counting while it is still sharper than the layer it marks.

3.3 How old the bottom of the snow is

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · BALLPARK

SCENE 33W

Brandt has the temperature string reading down to eighty-eight metres, which is where the spaces between the grains seal here. The surface adds about a third of a metre of snow a year.

GUIDE 58W

Four numbers, and two belong to other questions. The ice figure, and how heavy solid ice is. Ask of each whether it describes snow as it lies now. Up here the layers have not been squashed thinner by the flow, so this is one simple division. Deeper down the same sum has to be done bit by bit.

ASK 12W

How old is the snow at the depth where the spaces seal?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a depth divided by what one year adds gives a number of years

VERDICT 79W

Turning a depth into an age is division. How deep it is, divided by how much one year adds. Up here the layers have not been squashed thinner yet, so one division does it. Deeper down the same sum has to be done bit by bit, because each year is thinner than the one above it. Nearly three centuries of snow sit above the sealing depth here, and every one of those years is still open to the air.

TAKEAWAY 12W

A depth becomes an age only through the rate that filled it.

3.4 A year, or a summer that melted

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · BELT

SCENE 33W

Sections come out faster than they can be logged. A layer made over a year looks one way on the light table. A layer where the surface melted and froze again looks another.

GUIDE 37W

Two bins. A year of snow has coarse summer grains, fine winter snow, and bubbles all through it. A melt layer is clear and heavy with almost no bubbles, because water filled the spaces before it froze.

ASK 12W

Send each section to the bin that says what made the layer.

BEHIND THE BUTTON 55W

Why it matters so much. Counting layers is how the core is dated. A melt layer counted as a year makes everything below it too old, all the way down.

What melt does as well. Water running down carries the chemistry with it, so a melt layer smudges the record on both sides of itself.

TAKES AS READ

snow builds up in layers and can melt and freeze again

VERDICT 140W

A year of snow leaves a pattern. Coarse crystals from the summer sun, fine packed snow through the winter, and air trapped as bubbles all through it. Melt leaves something different. The surface thaws, the water runs down into the spaces and freezes there, so the ice comes out clear, heavy and almost free of bubbles. Both are stripes in a core and only one of them is a year. Counting matters because it is the clock. A melt layer counted as a year makes every date below it too old. A year missed makes every date too young. And melt does something worse than miscount, since the water carries chemicals down with it and smudges the record either side. That is why a drilling site is chosen for how rarely it melts, not for how much snow it gets.

TAKEAWAY 13W

A layer counted wrong shifts every date below it, in the same direction.

3.5 Why the hole is on the dome

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CALLBACK · CHOICE

SCENE 35W

Pär Lindqvist, the drill engineer, has the hole down at two thousand metres. On the wall is a map with three abandoned sites, all of them on slopes. Vestri Dome is the high point here.

GUIDE 55W

Four things a dome might buy, and three of them are about cold, thick or flat. Ask of each whether it is about where the ice came from. An ice sheet spreads out from its high points. On a slope, the ice deep down arrived from somewhere else. Domes pay for that with thin layers.

ASK 14W

What does drilling on the dome buy that a site on a slope cannot?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

ice spreads outwards from the high points of an ice sheet

VERDICT 77W

An ice sheet spreads outwards from its high points, like very slow treacle. On a slope, the ice at a given depth came from somewhere uphill. So a core there mixes one place's weather with a long journey, and nothing afterwards can separate them. Under a dome the ice goes almost straight down, so the column under the drill is very nearly the snow that fell on it. Domes charge for that. High, cold, dry, thin layers.

TAKEAWAY 18W

A core is a column of what fell above it only where the ice has not travelled sideways.

Every dating method has a depth where it stops working, and knowing it is part of the date.

Turning a ratio into a temperature

PLAN CARD 68W

Friday, and the isotope readings run down to sixteen hundred metres. Tanabe has the numbers, and Halvorsen wants them in degrees for the summary. That means a conversion figure. Somebody measured that figure; it is not a law of nature. Today you turn the readings into a temperature change, and decide how firmly it can be stated. A figure taken on trust turns every degree into somebody's assumption.

BRIEFING 17W

The isotope profile is finished to 1,600 metres and nobody has agreed what it says in degrees.

WHAT YOU DECIDE 15W

Establish what the isotope record measures and how confidently it can be stated in degrees.

4.1 What the ratio says in degrees

COLD LABORATORY · A ROOM · BALLPARK

SCENE 27W

Tanabe has the isotope readings against depth. The stretch covering the last cold spell sits 1.8 units below today's value. The conversion figure is on the board.

GUIDE 52W

Four numbers, and two belong elsewhere. One is a figure for a different measurement. One is the years overlapping the mast. Ask of each whether this conversion uses it. And note where the site figure comes from. It is not a law of nature but a fit to this place's own weather.

ASK 9W

How much colder was that cold spell than today?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

colder air drops its heavier water first, so the snow that falls last is lighter

VERDICT 73W

Water carrying heavier atoms turns to snow first. Air that has cooled further has already lost more of them, so the last snow to fall is the lightest. Over a useful range the reading and the temperature move together in a straight line. How steep that line is belongs to this place, not to physics. So it is measured against the mast. Divide the shift by it, and the answer is in degrees.

TAKEAWAY 16W

A conversion figure turns a measurement into a claim, and it is measured rather than assumed.

4.2 How firmly the cooling can be stated

SURFACE & ACCUMULATION · SNOW STUDY HUT · A PERSON · CLOUD

SCENE 27W

Brandt has the mast record, the years that overlap it, and the scatter around the fitted line. The season summary needs a statement about that cold spell.

GUIDE 70W

The band is the cooling worked out from the core, with the uncertainty of the calibration on it. The summary wants to say the spell was more than two degrees colder. Two pieces of work are ready, and they do different things to the band: one moves it, one makes it narrower. Apply what you would fund, watching the middle and the width, then say whether the claim is cleared.

ASK 27W

Which work would let the summary say the spell was more than two degrees colder? Move the bars to show the middle and one sigma either side.

BEHIND THE BUTTON 152W

Where the width comes from. Nobody measured the temperature. It is worked out using a relationship, and that relationship is calibrated on the years where the mast record and the core overlap. The scatter of those years around the fitted line is the uncertainty, and it comes along with every temperature the relationship gives.

Moving the band is not narrowing it. Changing the correction slides the whole band sideways. It is just as wide as it was. More overlap years make the calibration tighter, and that is the only thing that makes the band narrower. If the edge of the band is what crosses two degrees, only the second kind of work can clear the claim.

Why the claim has a number in it. 'Colder than now' cannot be checked. 'More than two degrees colder' can. And it means the whole band has to sit past two degrees, not just its middle.

TAKES AS READ

a line fitted through measurements carries a spread as well as a middle

VERDICT 78W

The answer is a spread, not a number. The line was fitted through eleven years of overlap, and it carries the scatter of those eleven years. Saying the spell was more than two degrees colder is a claim about where the whole spread sits. So it is won by making the spread narrower. More overlap years is the only thing here that adds information. The correction moves the middle and leaves the spread as wide as it was.

TAKEAWAY 15W

Moving the middle and narrowing the spread are different actions, with different evidence behind them.

4.3 The temperature of what, and where

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CHOICE

SCENE 34W

Jonas Aalto, who keeps the records, is drafting the summary. He has written that the isotopes give the temperature of the last cold spell. The drilling lead wants that narrowed before it goes out.

GUIDE 57W

Four things the isotope record might be about. Ask of each where and when the reading is actually set. It is fixed as the snow forms, up in the cloud, in the seasons that bring most of the snow. Getting from there to a ground temperature takes another step. Two of these options are separate measurements entirely.

ASK 8W

What does the isotope record most directly record?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

snow forms in a cloud and falls to the ground some way below it

VERDICT 68W

The isotope ratio is set as water vapour turns into snow. So it records what the cloud was like, weighted to the seasons that bring most of the snow. Getting from there to the ground needs the link between cloud and ground. That is what the site's own figure supplies. The sea the water came off leaves a small mark. The ice's own temperature is a different measurement.

TAKEAWAY 13W

A stand-in measures one specific thing, and the sentence has to say which.

4.4 What makes one year different from the next

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CALLBACK · PROTOCOL

SCENE 33W

A metre of core from sixty metres down is on the light table. Ines Okonkwo, the chronology lead, has four things marked on it and wants each one explained before it is cut.

GUIDE 50W

Four things marked on one metre of core, and four things that made them. Pair them by asking what each is made of. Crystals? Acid? Dust grains? Bubbles? Then ask which season it belongs to. No one marker lasts the whole core, so a year is where several signals agree.

ASK 11W

Match each thing in the ice to what put it there.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

snow that falls in summer sits in different conditions from snow that falls in winter

VERDICT 75W

No single marker survives all the way down a core. So a year is counted where several different signals line up. Coarse crystals grow on a summer surface left out in the open. A spike in electricity is acid, and volcanic acid arrives in one season. Dust blows in from far away and peaks with the winds there. The banding is bubbles changing as each layer was squashed. Three of four agreeing is a year.

TAKEAWAY 15W

A year is counted from several signals that agree, not from one that looks convincing.

END OF THE DAY 16W

A proxy becomes a number through a calibration, and the calibration is part of the result.

Air that stopped moving

BEFORE THE DAY — HUNT

Six core boxes, and four are labelled

Every box has to be found and matched to a depth before the plane comes. A box with no depth on it is ice nobody can date, and there is no second chance at this hole.

PLAN CARD 80W

Saturday. Samuel Adeyemi has carbon dioxide from four depths and a problem with all of them. The air in a bubble stopped moving centuries after the snow around it fell. At a site this dry, that gap is the biggest correction in the record. Today you work out how big the gap is here, and why the gas record is smoother than the ice one. Get it wrong and a gas record lands centuries from the climate it belongs to.

BRIEFING 16W

The gas record and the ice record from the same depth are not the same age.

WHAT YOU DECIDE 17W

Establish what a bubble is, when it sealed, and why its air is younger than its ice.

5.1 How much younger the air is

TRAPPED GAS · GAS LABORATORY · A ROOM · BALLPARK

SCENE 29W

The spaces between the grains seal at eighty-eight metres here. Above that, air moves freely up to the surface. Adeyemi has the yearly build-up on the board beside it.

GUIDE 66W

Four numbers, and one of them is the same year written as snow rather than as ice. Ask of each whether the units match the others. The air keeps mixing with the sky until the spaces seal. So the bubble is as old as the sealing, and the ice is as old as the snow. Get it wrong and the gas is plotted centuries too early.

ASK 11W

How much younger is the air than the ice around it?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

air moves through the connected spaces in the snow until they close off

VERDICT 79W

Air moves through the open spaces in the snow. It stays in touch with the sky until those spaces close. So the air in a bubble is about as old as the moment it was sealed. The ice around it is as old as the snow. The gap is how long that snow took to get down there. Cold dry places take centuries. Wet coastal ones take decades. Forget it, and a gas record is plotted centuries too early.

TAKEAWAY 18W

A bubble is sealed at a depth, and the ice around it arrived at that depth long before.

5.2 From snowfall to sealed bubble

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · CHAIN

SCENE 25W

Lindqvist has the borehole's own record on the wall. Five stages, from a snowflake landing to air being shut away, with a measurement against each.

GUIDE 55W

Put the five stages in the order a snowflake really goes through them, from landing to being shut away. Then name the stage that decides how much older the ice is than the air inside it. Each stage has a measurement against it, so read those instead of picking the stage that sounds most important.

ASK 14W

Which stage decides how much older the ice is than the air inside it?

BEHIND THE BUTTON 163W

Why the ice and its air are different ages. New snow is open to the air above it, and it stays open while it squashes down. Air keeps moving in and out of that layer for a very long time. Only when the tiny gaps finally pinch shut is the air trapped — and by then the ice around it has been buried for a long time.

What decides how big the gap is. How deep the gaps pinch shut, and how long the snow takes to get there. Both depend on how cold it is and how much snow falls each year. A cold, dry place has a gap of hundreds of years. A warmer, snowier place has much less.

Why it matters. Every comparison between the air record and the ice record has this gap in it. Ignore it and you can get the order of two events wrong, and the order is what tells you which one caused the other.

TAKES AS READ

something that passes through several stages is limited by one of them

VERDICT 73W

Every stage moves something, and only one fixes the size of the gap. Air swaps with the sky all the way down through the snow. So its age hardly changes, while the ice around it gets older year by year. The moment the spaces seal is the moment that swapping stops. Whatever the ice's age is at that depth is the gap. Mixing keeps the air young, and it is worth a decade.

TAKEAWAY 16W

In a chain of stages, one of them sets the answer and the rest follow it.

5.3 Why the gas record has no sharp edges

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CHOICE

SCENE 32W

Okonkwo has both records from the same depths on one plot. The ice record has features a few years wide. The gas record has nothing in it narrower than a few decades.

GUIDE 53W

Four explanations for a record with no sharp edges. Ask of each whether it is about this place or about the laboratory. Cutting smaller pieces would help one of them a little. The spaces do not all seal at once, so the air trapped in one layer is a mixture of many years.

ASK 14W

Why is the gas record smoother than the ice record from the same depths?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a measurement that averages over a stretch cannot show anything shorter than it

VERDICT 74W

Air stays in touch with the sky all the way through the open snow, and the spaces do not all close at once. They close across a stretch of depth, so the bubbles sealed in one layer hold a mixture covering every year that stretch represents. That smoothing belongs to the site. Somewhere with heavy snowfall seals in a decade and keeps sharp features. Here it takes several. Cutting smaller samples cannot undo it.

TAKEAWAY 13W

A record made by averaging cannot show a change faster than the averaging.

5.4 How thick a year is here

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · CALLBACK · BALLPARK

SCENE 27W

Brandt has the pit wall photographed. Last summer's surface is marked thirty-two centimetres down. The snow above it weighs about a third of what solid ice would.

GUIDE 65W

Five numbers, and two of them belong elsewhere. How heavy water is, and the days of stake readings. Ask of each whether this sum needs it. Snow and ice are the same stuff at different weights, so a thickness of snow is not a thickness of record. Only ice lets two sites be compared. This one gets about a tenth of what the coast gets.

ASK 9W

How much ice will this year's snow turn into?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

snow turns into ice by being squashed under the snow that falls on top of it · density is how much a given amount of something weighs

VERDICT 74W

Snow and ice are the same stuff at different weights, so a thickness of snow is not a thickness of record. Scale it by how heavy the snow is against solid ice, and you get what it will become once the air is squeezed out. That is the only form in which two places can be compared. Here a year is about twelve centimetres of ice. A site by the coast gets a metre.

TAKEAWAY 15W

A year's snowfall means nothing until it is written as the ice it will become.

END OF THE DAY 19W

Two things in the same piece of ice can be different ages, and the difference has to be calculated.

The year that is written everywhere

PLAN CARD 65W

Monday. The chemistry has found a sulphur spike with glass fragments in it. If it is the eruption Okonkwo thinks it is, the deep section gets its first fixed date. Adeyemi wants to know what that does for the gas record. The forecast gives Brandt one clear day for three jobs. Today you name the spike, and choose what gets done before the wind returns.

BRIEFING 14W

A sulphate spike at 1,642 metres could be the anchor the deep section needs.

WHAT YOU DECIDE 19W

Use a dated eruption to tie this core to another one, and know what that tie does not fix.

6.1 The spike at 1,642 metres

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · DIAGNOSIS

SCENE 43W

The chemistry has found sulphate, which is volcanic sulphur laid down in a thin band within a year or two of an eruption. Two eruptions are of about the same age. Glass fragments came from the same centimetre. Okonkwo has the layer count.

GUIDE 55W

Five readings, and only one of them owes nothing to the clock. Ask of each whether it could tell two eruptions of nearly the same age apart. The modelled age cannot: its own doubt is wider than the gap. Sea salt does not arrive with volcanic glass. And the layer count stopped far higher up.

ASK 7W

Name which eruption the layer came from.

BEHIND THE BUTTON 156W

Why the quiet readings matter. It is easy to build a story around the number that is out of range. But most wrong explanations fail on something that did not move. If a cause would have pushed a second reading up and that reading is where it should be, the cause is not what happened, however well it explains the first one.

How to use the candidates. Take each one and ask what the panel would look like if it were true. Then compare that with the panel in front of you. Doing it in that order stops you fitting the story to the numbers, which is easy to do without noticing.

Why there is only one answer. More than one candidate will fit part of the panel, and that is on purpose. Only one fits every part of it. If two still look possible, find the reading they disagree about and let that reading choose.

TAKES AS READ

volcanic glass carries the chemistry of the volcano that threw it out

VERDICT 70W

The glass chemistry is the only reading here that does not depend on the timescale. It is also the one that tells the two apart, because glass carries the fingerprint of the magma it came from. The modelled age cannot choose between two eruptions ninety years apart. Its own doubt is wider than that. Sea salt does not arrive with volcanic glass. And the layer count stopped far above this.

TAKEAWAY 17W

A layer is identified by what is in it, not by the date that would be convenient.

6.2 What a dated ash layer fixes

TRAPPED GAS · GAS LABORATORY · A PERSON · CHOICE

SCENE 24W

Adeyemi has the eruption's date, the depth it sits at, and the carbon dioxide measured from the bubbles in the same piece of core.

GUIDE 56W

Four things a dated ash layer might fix. Ask of each which of the two clocks in the core it belongs to. Ash lands on the surface and is buried with the snow. The air beside it was sealed much later, further down. So the layer fixes one clock and leaves the other to a sum.

ASK 6W

What does the dated layer fix?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

volcanic ash falls out of the sky within a year or two of the eruption

VERDICT 79W

Ash falls onto the surface and is buried with the snow, so it dates the ice it sits in. The air in the bubbles beside it was sealed centuries later, further down. It is that much younger. So a dated layer fixes one of the two clocks in the core, and leaves the other to be worked out. That is why the age gap has to be found before a gas record and an ice record can be compared.

TAKEAWAY 17W

A dated layer dates the material it is in, and anything else has to be worked out.

6.3 One clear day, three jobs

SURFACE & ACCUMULATION · SNOW STUDY HUT · A ROOM · TRIAGE

SCENE 27W

Brandt has a forecast with one clear day in it and then four of blowing snow. Three jobs are outstanding, and each takes most of a day.

GUIDE 59W

Four jobs, each a day's work. Ask of each whether it could be done in four days of blowing snow. Then ask whether the storm changes what it measures. The pit is by the hut and the mast is on a fixed line. One of these runs two kilometres out, and drifting snow rearranges the very thing it measures.

ASK 6W

Which job takes the clear day?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

a measurement that cannot be repeated should be taken while it can be

VERDICT 78W

Two of these can be done in bad weather and one cannot. The pit is beside the hut and the mast is a short walk along a fixed line. Both survive four days of blowing snow. The stake array runs two kilometres out and has to be seen to be read. Worse, blowing snow moves the very surface the stakes are measuring, so a reading taken afterwards measures something else. The broken core can be redrilled any day.

TAKEAWAY 13W

Order the work by what becomes impossible, not by what feels most important.

6.4 What the saw is cutting for

COLD LABORATORY · A ROOM · SPOT

SCENE 26W

Sections come to the saw with a cutting list. The list is set by whichever job has priority, and that changes as the season goes on.

GUIDE 47W

Cut what the list asks for and leave the rest in the rack. The priority on the board changes during the day and nobody says so. What counts is the sections either side of a change, because a section cut for the wrong job cannot be uncut.

ASK 12W

Cut to the priority on the board, and keep checking the board.

BEHIND THE BUTTON 64W

Why the priority changes. Early the season is about dating the core. When the gas laboratory books its slot it becomes anything for gas. Before the aircraft it becomes anything that has to fly out.

Why this one cannot be repeated. Almost everything else here can be done again with more time. A cut cannot. The ice at that depth cannot be drilled twice.

TAKES AS READ

a core section can only be cut up once

VERDICT 115W

Three priorities run across the day and each wants different sections. Anything for the dating work, then anything for gas, then anything leaving on the aircraft. A section can match two at once, which is what makes a change cost something. What the panel scores is the sections either side of a change. Most of the rack is wanted by neither priority, so leaving those alone proves nothing. This is the sharpest case in the camp, because a cut cannot be undone. Everything else can be repeated with more fuel or more time. A section cut for isotopes is gone from the gas laboratory for good, and nobody is drilling that depth again this season.

TAKEAWAY 14W

A rule that has changed costs most where the work cannot be done again.

6.5 The run through the brittle zone

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CALLBACK · SEQUENCE

SCENE 32W

The barrel is at seven hundred and eighty metres, in the depth band where the ice comes up cracked. The winch, the fluid and the storage are all set for ordinary ice.

GUIDE 58W

All five steps will happen, so ask what each one buys the next. Down in that band the bubbles are squeezed harder than the surface can hold. Ask of each step whether it softens the shock, lets the pressure ease, or saves something about to be lost. One of them throws away the depths if it comes first.

ASK 13W

Put the steps in the order that brings a cracking section up whole.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

air trapped deep in the ice is squeezed to the pressure of the ice around it

VERDICT 74W

Deep down, the bubbles are squeezed far harder than the air at the surface. Bring that ice up quickly and it cracks itself apart. Each step here buys the next one. A slow lift is less of a shock. Resting at hole temperature lets the pressure even out. Writing down the breaks fixes which depth each piece came from, while somebody still knows. Cutting first destroys that, and cuts ice that is still moving.

TAKEAWAY 12W

Ice brought up badly is a shorter record, not a cheaper one.

END OF THE DAY 17W

A dated event in two records fixes them to each other at that moment, and nowhere else.

Two records, side by side

BEFORE THE DAY — CANVASS

What the last team left behind here

A camp stood on this spot four years ago, and what they buried changes what our samples mean. Ask around the camp until enough of the old hands have said what was here and where it went.

PLAN CARD 76W

Wednesday, and both records are on one screen for the first time. They differ by about half a unit through the cold spell, which is more than either team has claimed as its doubt. Halvorsen has a draft paragraph written. Today you work out which parts of the comparison stand on their own, and what half a unit is worth in degrees. Two records that share a method are one record with two names on it.

BRIEFING 14W

The Skarv record and this one differ by more than either group's stated uncertainty.

WHAT YOU DECIDE 16W

Put both records on one plot and find out how much of the comparison is independent.

7.1 What both records rest on

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · TRACE

SCENE 31W

Five pieces of the comparison are on Aalto's screen, each worked out from something else. The chronology team wants to know which would still stand if the flow model were wrong.

GUIDE 51W

Open each piece of the comparison to see what it was worked out from. Keep the ones that would still stand if the flow model were wrong, and untick the ones that go through it. Then name what those share. A piece that survives is worth more than its number suggests.

ASK 12W

Which parts of this comparison would survive the flow model being wrong?

BEHIND THE BUTTON 132W

What the flow model does. It turns a depth in the ice into an age, by describing how the ice has squashed and moved since it fell. Nearly every dated quantity has been through it, which makes it the most important assumption in the building.

Why some things survive it. Anything measured against a fixed marker — an ash layer, a known event, a layer somebody counted directly — is a fact about the ice rather than about the model. Those hold whatever the model does, and they are how the model gets tested.

Why ask it this way round. Nobody thinks the flow model is simply wrong. The question is which conclusions depend on it, so that when it is revised the team knows exactly what has to be done again.

TAKES AS READ

two results worked out from one assumption fail together if that assumption fails

VERDICT 68W

Both teams counted layers as far as they could. Then they handed the rest to the same flow model. Below the counts, the two age scales are not two estimates that agree. They are one calculation done twice. What survives the model being wrong is the counted part, which is arithmetic on measured layers, and the ash layer, whose date came from a volcano rather than from ice.

TAKEAWAY 16W

Two numbers that agree because they came through the same step have not checked each other.

7.2 How much more carbon dioxide

TRAPPED GAS · GAS LABORATORY · A PERSON · BALLPARK

SCENE 28W

Adeyemi has carbon dioxide at 190 parts per million through the cold spell, and 280 at the top of the core. The station wants the change stated plainly.

GUIDE 56W

Four numbers here, and one of them belongs to a different measurement. Another is the gap between the two, worked out for you already. Ask which number a percentage is measured against. It is always the value you started from. Pick the other one and the answer comes out too small, and it still looks fine.

ASK 10W

How much bigger is 280 than 190, as a percentage?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

parts per million counts how many there are out of every million

VERDICT 73W

Ninety parts per million sounds small on its own. Everything depends on what it started from. Against a hundred and ninety it is a rise of nearly half. That is the number that matters. What the air does with heat depends on how much of this gas is in it, not on how many parts got added. State a change without its starting point, and a big change reads as a small one.

TAKEAWAY 13W

A rise means most when it is stated against what it started from.

7.3 Half a unit, in degrees

COLD LABORATORY · A ROOM · CHOICE

SCENE 29W

The two records differ by about half a unit through the cold spell. Tanabe has the site's own figure, and a draft that calls that difference a different climate.

GUIDE 58W

Four readings of half a unit, and one refuses to convert at all. Ask of each whether it uses this site's own figure, and which way round. Multiplying where you should divide gives an answer far too big. And note why converting matters. In units it sounds like rounding. In degrees it is a quarter of the headline.

ASK 13W

What is half a unit worth, in the terms the draft paragraph uses?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a difference in units becomes a difference in degrees through the site's own figure

VERDICT 70W

This site's figure is 0.67 units for every degree. Half a unit divided by that is about three quarters of a degree. That is the whole point of converting. In units the difference sounds like a rounding error. In degrees it is a quarter of the cooling the season means to report. Whether it is real climate is a separate argument, and nobody can have it in two different units.

TAKEAWAY 17W

A difference has to be put into the units of the claim before anybody can judge it.

7.4 The metre with no label

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · CALLBACK · CHOICE

SCENE 32W

A section has come to the line with its depth mark broken. Holt can place it to within about four metres from the run sheet. The isotope team has asked for it.

GUIDE 62W

Four options for a section nobody can place better than four metres. Ask of each what the depth is for. It is what turns a measurement into a statement about a year, and four metres here is about thirty years. Some work never uses depth at all. And the doubt has to stay with the sample, or somebody later will not know.

ASK 7W

What should the section be used for?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a measurement belongs to the depth it came from, and to no other

VERDICT 80W

A depth is what turns a measurement into a statement about a year. Four metres here is thirty-odd years, so an isotope value off this section cannot be put anywhere in the record. The ice is still perfectly good for anything a depth does not enter. How heavy it is. What the crystals look like. How the drill is performing. What matters is that the doubt travels with the sample. The danger is somebody later who never hears about it.

TAKEAWAY 12W

A sample with no depth is a measurement of nothing in particular.

END OF THE DAY 11W

Agreement between two records is worth what their independence is worth.

The offset that was not climate

BEFORE THE DAY — EVADE

Keep clear of the drill floor while the winch runs

The cable is under load and nobody on that floor can hear you. Walking through while the winch is running is how somebody gets hurt. Go round, and get to the trench the long way.

PLAN CARD 67W

Thursday. Adeyemi has spent the night on the two teams' age scales. Skarv gets four times the snow this station does, so its gap between gas and ice is a fraction of ours. Today you predict what the difference becomes once that is corrected, do it, and check it against the one date neither team worked out. If the correction explains the difference, the two records agree.

BRIEFING 14W

The two records may differ because one dates ice and the other dates gas.

WHAT YOU DECIDE 14W

Test whether the age difference accounts for the disagreement before it is called one.

8.1 What the correction leaves behind

TRAPPED GAS · GAS LABORATORY · A ROOM · VERIFY

SCENE 28W

Adeyemi has the age gap for both sites, and the means to re-plot both gas records on their own clocks. The ash layer is dated by neither calculation.

GUIDE 52W

Write down what you expect the difference to become once both gas records are on their own clocks. Then re-plot them and see. The ash layer is dated by neither calculation, so it is the check that can tell you whether the correction did what you predicted rather than what you hoped.

ASK 15W

What is left of the difference once both gas records are on their own clocks?

BEHIND THE BUTTON 144W

What the correction does. At each site the trapped air is younger than the ice around it, and by a different amount at each. Putting each gas record on its own clock takes that difference out. Whatever is left is a real difference between the two places.

Why write the prediction first. If you re-plot and then decide what you expected, anything left over can be explained away afterwards. Committing to a number first is what lets the result surprise you, and being able to be surprised is what makes it evidence.

Why the ash layer works as a check. Ash from one eruption lands at both sites within days of each other, and its date does not come from either calculation. If the correction is right, the ash lines up in both records. If it does not, something in the correction is wrong.

TAKES AS READ

a correction has to be predicted before it is applied, or it cannot be wrong

VERDICT 67W

Air keeps mixing with the sky until the snow seals, so the gas is younger than the ice around it. How much younger depends on the sealing depth and how fast snow arrives. Skarv gets four times the snow, so its gap is a fraction of this station's. Plot both on ice clocks and one curve shifts much further. That looks like climate and it is arithmetic.

TAKEAWAY 14W

A prediction made before the correction is the only thing that can test it.

8.2 The clock that does not need layers

CHRONOLOGY & THE CORE LINE · CORE LINE · A PERSON · BALLPARK

SCENE 24W

Okonkwo has a sample from the deep section. A quarter of the radioactive atoms are left, and the half-life is written on the bench.

GUIDE 54W

Four numbers, and two of them are counts of halvings. Only one of those is right for a quarter left. Ask of each what it counts. Each halving takes the same length of time, so the question is how many of them have gone by. The ice thickness has nothing to do with it.

ASK 5W

How old is the sample?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a radioactive atom halves in amount every half-life, whatever else happens to it

VERDICT 62W

A radioactive atom decays at a rate nothing in the ice can change. Not temperature. Not pressure. Not flow. Every half-life, half of what is left goes. So a quarter left means two halvings have gone by, which is two half-lives of time. That is a date owing nothing to the flow model, which is what makes it worth having down here.

TAKEAWAY 17W

Where counting has run out, a clock that runs on its own is the only date left.

8.3 What is left to disagree about

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CHOICE

SCENE 27W

Both records are re-plotted, the ash layer lines up in both, and 0.15 of a unit of difference is left. Both teams quote doubt bigger than that.

GUIDE 60W

Four readings of what is left after the correction. Ask of each whether the difference left over is bigger or smaller than the doubt each team quotes. Then ask what a correction does to a result. This one was worked out before it was applied, and checked against a date from neither team. Both records were on the wrong clock.

ASK 6W

What does the corrected comparison support?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a difference smaller than the doubt on either measurement is not a measured difference

VERDICT 67W

What is left after the correction is smaller than either team's own doubt. So there is no measured difference left to explain. That is a result, rather than the absence of one. Two cores, drilled by different people and put on clocks that match, record the same cooling. The correction does not weaken that. It was predicted first, and then tested against a date neither team produced.

TAKEAWAY 15W

Explaining a disagreement away is a result, and it is the result that gets published.

8.4 Clean until it was handled

COLD LABORATORY · A ROOM · CALLBACK · CHOICE

SCENE 46W

Elena Cruz has a salt result four times the run before it. The blank run before it was clean, and the blank after it is high. The outer centimetre was trimmed off as it should be, and the same year in the second core is ordinary.

GUIDE 59W

Four options, and the clean readings do the work. The blank before was fine and the blank after is high. The trim was done properly. The same year in the second core is ordinary. Ask of each option how many of those it fits. What names the source is not the loud number but which quiet ones stayed quiet.

ASK 19W

The blank before the sample was clean, and the blank after it is high. Where did the salt join?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a blank goes through the whole process with no sample in it, so it measures the process

VERDICT 74W

The blank before the sample is normal, and the blank after it is high. That puts the salt inside the run rather than inside the ice. The trim was done properly, so the outside of the section is out. The second core holds the same year with no salt in it, so that year's weather is out too. What names the source is not the loud number. It is which quiet ones stayed quiet.

TAKEAWAY 15W

A blank measures the process, so anything it shows was never evidence about the ice.

A disagreement between two records is a claim about both of them, including how they were dated.

The night the generator stopped

PLAN CARD 73W

Saturday. The number two generator stopped at some point in the night. The core store came back at minus 19, and the log has an entry for the whole night, signed. Anneke de Vries needs the drill and the store back in a fixed order. Today you decide which signatures are worth checking, and what gets restarted first. Ice that warmed and was signed for as cold is worse than ice that warmed.

BRIEFING 15W

The store was without power for six hours and the log says nothing about it.

WHAT YOU DECIDE 14W

Work out what the core store record can still prove after a power failure.

9.1 Six signatures and two hours

COLD LABORATORY · A ROOM · ATTEST

SCENE 24W

Six lines on the night sheet, all signed. Tanabe has two hours before the flight list closes, and can properly check two of them.

GUIDE 56W

Four signed lines, two hours, and two of them can be checked properly. Open each claim and read what backs it now. A signature says somebody took responsibility, not that anybody went and looked. Hold what the evidence does not support, and spend the two hours where being wrong would matter after the flight has gone.

ASK 8W

Which two claims should the two hours check?

BEHIND THE BUTTON 126W

Why the deadline is the pressure. Once the flight list closes, anything unchecked stays unchecked until next year, and the samples are already off the ice. A check next season is a check on something nobody can go back to.

What a night sheet is. A record of what was done on shift, signed at the end of it. It is written by tired people at the end of a working night, and it is meant to pass on what happened rather than to prove it. That is why some lines are firmer than others.

How to choose between two doubtful lines. Not by which is least likely to be true. By which one, if it is wrong, changes what somebody does with the material later.

TAKES AS READ

a signature records that somebody believed something when they signed

VERDICT 69W

A signature says somebody believed something at the time. The store reading of minus nineteen says that belief was wrong for part of the night. Two of these claims matter and are carried by nothing but a signature. The store, and the gas samples. Both decide whether measurements on that ice can be defended. The mast keeps its own record. The food and the fuel are the camp's problem.

TAKEAWAY 16W

A signed record is evidence that a claim was made, not evidence that it was true.

9.2 What comes back first

DRILLING & RECOVERY · DRILL TRENCH · A PERSON · TRIAGE

SCENE 26W

Power is back on one generator. Three teams each want their own system first, and Lindqvist is arguing hardest. One generator will not carry them all.

GUIDE 54W

Four systems, and three of them are waiting. Ask of each whether the delay costs time or costs evidence. Warm ice starts losing gas out of its cut faces, and nothing gets it back. A lost drilling day is a day. Queued samples are as good tomorrow. Sort by what is still being lost.

ASK 4W

What gets restarted first?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

something already being lost is different from something merely waiting

VERDICT 72W

Three of these are waiting and one is going bad. Ice above minus twenty starts losing gas out of the bubbles near its cut faces. That loss cannot be undone or measured around, so every hour is more core that can never answer a gas question. A lost drilling day is a day. The queued samples are as good tomorrow as today. A report sent six hours late is a report sent.

TAKEAWAY 17W

Restart in the order of what is still being lost, not in the order of who asked.

9.3 The depth the counting runs out

CHRONOLOGY & THE CORE LINE · CORE LINE · A ROOM · SWEEP

SCENE 54W

Snow piles on snow. Each year makes a layer. The weight above squashes the older layers thinner. The scanner can tell two bands apart if they are three millimetres apart. Any closer and it sees one band. Okonkwo needs the depth where a year gets that thin, because past it nobody can count years.

GUIDE 55W

Drag the depth. The readout is how thick one year of ice is there, in millimetres. Only the depths you stop at get drawn. You are not looking for a high point. You want the depth where the thickness drops to three millimetres, which is the scanner's limit. Find that depth, mark it, and commit.

ASK 19W

The line separates bands three millimetres apart and no closer. At what depth does a year get that thin?

BEHIND THE BUTTON 175W

Why the layers get thinner. Ice is heavy. The ice above presses down on the ice below, and an ice sheet also spreads out sideways under its own weight. Both of those stretch a buried layer thinner. Near the top, one year is a few centimetres of ice. Two kilometres down, the same snowfall may be a few millimetres.

What is lost below the limit. Above the crossing, one year is a band you can point at, and the age of the ice is a count. Below it, two years land inside one scanner width, and the count stops. Ages then come from a model of how the ice flows. A model is a good guess with an error bar, not a tally.

Why this depth is measured in every core. How fast the layers thin depends on how the ice flows there, and no two places flow the same. So the limit sits at a different depth in every core, and it has to be found again each time before anybody publishes an age.

TAKES AS READ

layers are stretched thinner as the ice underneath them spreads outwards

VERDICT 73W

Ice spreads outwards under its own weight, so every layer is stretched thinner as it sinks. The deeper it goes, the faster that happens. At the surface a year here is about twelve centimetres. Two kilometres down the same year is a few millimetres. Where the layers close to within what the line can separate, the counting stops. Not because the years have gone, but because two of them cannot be told apart.

TAKEAWAY 13W

A count ends where the instrument stops telling one year from the next.

9.4 One year proves nothing

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CALLBACK · BALLPARK

SCENE 29W

There are thirty years of yearly temperatures from the top of the core. The summary wants one number, and one year can swing more than a degree either way.

GUIDE 53W

Four numbers, and two belong to other questions. How much one year swings, and the years overlapping the mast. Ask which two an average needs. And note what an average does not do. It does not make one year's swing go away. It only says where the middle of thirty of them sits.

ASK 9W

What is the average temperature across the thirty years?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

averaging many years lets the random part of each one cancel out

VERDICT 72W

Every year comes out a little warmer or colder than the run of them, because weather is noisy. Add thirty years and divide by thirty, and those swings cancel out. What is left is much closer to the climate of the period than any one year is. It is also why nobody should read a single warm year as a change. One year here swings more than the difference being argued about.

TAKEAWAY 14W

One year is weather. An average of many years is closer to a climate.

END OF THE DAY 17W

A record is a claim about a condition, and the two can come apart in a night.

What is known, and how well

BEFORE THE DAY — TAG

Catch Halvorsen before the plane loads

Halvorsen, the chief scientist, has the only copy of the depth log for the last eleven metres. Once the plane is loaded it goes, and a core with no log is a core nobody can put in order.

PLAN CARD 66W

Monday, the last working day, and the aircraft comes on Wednesday. The report is going out. Halvorsen wants the season's own account of itself. What is settled, what rests on a model, and what the next team does first. Today you sort what was measured from what was assumed, and say what the shortest event this ice could show is. Nobody comes back for four years.

BRIEFING 17W

The last day is spent deciding what the record supports and what the next season must do.

WHAT YOU DECIDE 14W

State what the season established, at what precision, and what it could not reach.

10.1 The sentence the record will carry

RECORDS & REPORTING · SCIENCE MODULE · A ROOM · CHOICE

SCENE 28W

Four versions of the season's headline claim are on Aalto's screen. All of them are true in some sense. One of them is what the evidence actually reaches.

GUIDE 55W

Four sentences, all true in some way. Ask of each what it claims beyond what was measured here. One calls the two records independent, and below the counts they share a model. One rounds until the doubt vanishes. One is a claim about every record at this latitude, and nobody here has read them all.

ASK 7W

Which sentence does this season's evidence reach?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a claim is supported when the evidence would have come out differently if it were false

VERDICT 64W

The first sentence says what was measured, over what stretch, and which part rests on something nobody measured here. The second calls the two records independent. The shared flow model makes that untrue below the counts. The third rounds until the doubt vanishes into the number. The fourth is a claim about every other record at this latitude, and nobody here has read them.

TAKEAWAY 15W

An honest claim says what was measured, over what period, and with how much doubt.

10.2 What rests on what

SURFACE & ACCUMULATION · SNOW STUDY HUT · A PERSON · PROTOCOL

SCENE 26W

Brandt has the season's four headline numbers up on the hut wall. Beside each one is an empty column. Nobody has filled any of it in.

GUIDE 58W

Four headline numbers, and four descriptions of where each came from. Pair them by asking how many steps sit between the instrument and the number. One is measured at the surface and checkable next year. One goes through a line fitted to eleven years. One is counted, then modelled. And one is solid, attached to a modelled date.

ASK 10W

Match each headline number to what it actually rests on.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a number is only as strong as the weakest step between the ice and it

VERDICT 77W

The four numbers are not the same kind of thing, and the report has to say so. The snowfall is a direct measurement of the surface, and next year's stakes can check it. The cooling is a measurement passed through a conversion, so it carries that conversion's doubt. The age is counted where counting worked and modelled where it did not. The gas amount is well measured, and it hangs on a depth whose age is modelled.

TAKEAWAY 20W

Every number in a record of the past belongs to a class, and the class is part of the number.

10.3 The shortest thing this ice can show

COLD LABORATORY · A ROOM · BALLPARK

SCENE 36W

Tanabe has how far water moves through the snow before it seals, and the thickness of a year at the surface. The next team wants to know whether one warm summer could ever be found here.

GUIDE 51W

Four numbers, and one of them is a year deep down rather than at the surface. Ask which thickness the question is about. Water moving through the snow blurs the layers together, and the answer is how many years fit inside that blur. Up here it is barely more than one.

ASK 7W

How many years get blurred together here?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

water moving through the snow blurs nearby layers into each other

VERDICT 81W

Water keeps moving between the grains until the snow seals, and it carries the isotope signal along with it. So anything thinner than the distance it travels gets averaged into its neighbours. Here that distance is about sixteen centimetres of ice, and a year is twelve. So the shortest thing this record can show is a bit over a year. One hot summer sits right at the edge. Deeper down, where a year is millimetres, it has no chance at all.

TAKEAWAY 18W

A record has a shortest event it can show, and the site sets it rather than the instrument.

10.4 The level that holds the hole open

DRILLING & RECOVERY · DRILL TRENCH · A ROOM · CALLBACK · HOLD

SCENE 37W

The hole is three kilometres deep and the ice around it is trying to close it. The fluid inside holds it open. The level in the tower has to stay where it belongs while the drill runs.

GUIDE 44W

Hold the level inside the band on the gauge. The band gets narrower as the hole gets deeper. Each run of the winch changes the balance and keeps changing it, so set the pump to match rather than waiting for the gauge to move.

ASK 8W

Hold the fluid level while the drill runs.

BEHIND THE BUTTON 57W

Why the hole needs holding open. Ice this deep is under enough pressure to flow, and it slowly squeezes a hole shut. Fluid pushing back from inside is what stops it.

Why too much is bad too. Push too hard and the wall cracks and swallows fluid. That costs the hole and spoils the record around it.

TAKES AS READ

a column of liquid pushes harder the deeper it is

VERDICT 139W

Ice three kilometres down is under enough pressure to flow, so a hole through it closes on its own. What holds it open is the fluid inside, pushing back. Too little and the walls creep in. Too much and the wall cracks and swallows fluid, which costs the hole and spoils the record near it. Everything the drilling does upsets that balance and keeps upsetting it. Pulling the drill up takes away what it was displacing, for as long as the winch runs. Chips come up with fluid on them. Cold fluid off the reel is heavier than what is in the hole and keeps sinking into it. So the pump is set to match, not tapped. The band gets narrower with depth because the gap between closing the hole and cracking it gets smaller the deeper you go.

TAKEAWAY 15W

A hole is held open by a balance, and somebody has to keep making it.

END OF THE DAY 11W

A record is worth what its limits are known to be.

THE CLOSING CARD

The season closed with the hole at 2,470 metres and a record that reaches, by counting, to about nine thousand nine hundred years, and by flow model to something under thirty thousand. The disagreement with Skarv turned out to be mostly the gas-ice age difference: four hundred and fifty-odd years at this site and a fraction of that at theirs, applied to both records and then checked against a tephra horizon that neither calculation produced. What was left of the difference was smaller than either group had claimed as its uncertainty, and the report said so in the hedged sentence rather than the confident one.

What it cost: a night of the core store above minus twenty, which took the gas measurements from four sections with it; a fortnight in which the comparison was argued about on two timescales that could not be compared; and the deep dating, which still rests on a flow model this station did not build. What is unfinished: below the counted section every age is modelled, the firn air profile was never re-measured, and the next team arrives to a question nobody here could answer — whether the same correction holds at four times the accumulation rate, which is the assumption the whole comparison now stands on.

Card text only — no options, boards or answers. 7,873 words of card prose across 41 stops. Generated from themes/icecore_ms by tools/make-cardbook.mjs.